

Sumit Sahu

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EXPERIENCE

Backend Intern

February 2026 – May 2026

OnArrival Bengaluru

- Boosted search API performance by integrating Redis caching, cutting database query load by 60% and reducing P95 latency from 800ms to under 150ms.
- Analyzed high-volume Elasticsearch logs and eliminated redundant entries via structured and level-based logging, cutting ingestion by 50% and improving query response times.
- Designed a Strategy Pattern-based data-sync pipeline to replicate traveler records to the Niyo client repository using WebClient fire-and-forget, eliminating partial-write failures during external service outages.

Backend Intern

July 2025 – September 2025

Vasitum Noida

- Designed and shipped 5+ production REST APIs using Spring Boot, reducing average endpoint response time by 35% through query optimisation and connection-pool tuning.
- Diagnosed and resolved 20+ critical production bugs via root-cause analysis of logs and thread dumps, improving platform uptime and reducing escalations.

Software Developer Intern

January 2025 – February 2025

Codes Mind Consultancy Services Bengaluru

- Designed and implemented structured prompts for training Large Language Models (LLMs) to optimize model accuracy in problem-solving and algorithmic tasks.
- Evaluated LLM responses for client Turing based on parameters including accuracy, correctness, and output efficiency.

PROJECTS

Advanced RAG Pipeline

[GitHub →](#)

Python, LangChain, Redis, BM25, Groq, Hybrid Search

- Built an end-to-end Retrieval-Augmented Generation (RAG) system for context-aware question answering across 300+ semantically chunked PDF documents.
- Improved retrieval relevance and answer accuracy by 70% using a hybrid retrieval architecture, RRF fusion, and Cross-Encoder reranking.
- Achieved sub-2s response generation with Redis-based caching, metadata-aware retrieval, and a scalable pipeline for enterprise knowledge retrieval.

Data Preprocessor Pipeline

[GitHub →](#)

Python, Pandas, NumPy, Scikit-learn

- Built an automated pipeline to clean and preprocess raw CSV datasets, handling missing values, duplicate removal, type inference, and outlier detection.
- Implemented modular transformation stages (normalization, encoding, feature selection) enabling plug-and-play integration with downstream ML models.
- Reduced manual data preparation time by ~80% for structured datasets, with configurable rules adaptable to different schema formats.

AI MathPad

[Live Demo →](#)

Python, React.js, FastAPI, LangChain

- Developed an AI-powered canvas interface that interprets hand-drawn mathematical and graphical problems, delivering accurate solutions without keyboard input.
- Replaced traditional keypad interaction with an interactive drawing canvas, cutting problem-input time by ~70% in user testing.

TECHNICAL SKILLS

Languages: Java, Python, SQL, HTML5

Frameworks & Tools: Spring Boot, FastAPI, LangChain, REST APIs, WebClient, Redis, Postman, Git, Claude, MCP

Databases: MySQL, MongoDB

ML & Data: Pandas, NumPy, Scikit-learn, RAG, Hybrid Search, Prompt Engineering

Concepts: Data Structures & Algorithms, OOP, Design Patterns, System Design

EDUCATION

Dr. APJ Abdul Kalam Technical University

November 2021 – July 2025

B.Tech in Computer Science – 73% Allahabad

Allahabad Public School

July 2020 – July 2021

Class 12th – 80% Allahabad

ACHIEVEMENTS

LeetCode: Solved 300+ problems; ranked in the top 20% globally in a biweekly contest.

Hackathon Mobilothon 2024: Shortlisted among 50,000+ participants organised by Volkswagen.

Hackathon Vultr (GeeksforGeeks): Shortlisted among 20,000+ participants.